

Infinitely many affine-inequivalent quadratic PCF maps

A disproof of the first assertion of conjecture 00000007755

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Correction to PR #826. The previous submission used $(X - a)^2 + a$. Although every member is PCF, these polynomials all lie in the affine-conjugacy class of X^2 . That family does not give distinct parameters in the intended moduli space. We replace it with a construction of infinitely many PCF maps in the normalized quadratic family $F_c(X) = X^2 + c$.

Put $K = \overline{\mathbb{Q}}$. A polynomial over K is PCF if every root of its formal derivative in K has a preperiodic forward orbit. The only critical point of F_c is 0, because $F'_c(X) = 2X$.

Theorem 1. *There are infinitely many parameters $c \in K$ for which F_c is PCF. The corresponding maps are pairwise inequivalent under affine conjugacy. Consequently the finiteness assertion of conjecture 00000007755 is false already for degree 2, even in the space of affine-conjugacy classes.*

Proof. Work first in $K[T]$. Define the critical-orbit polynomials by

$$P_0(T) = 0, \quad P_{n+1}(T) = P_n(T)^2 + T.$$

Induction gives $P_n(c) = F_c^{\circ n}(0)$. Define a second sequence by

$$Q_0(T) = 1, \quad Q_{n+1}(T) = TQ_n(T)^2 + 1.$$

Another induction shows $P_{n+1}(T) = TQ_n(T)$. For every $n \geq 1$, $Q_n(0) = 1$ and the coefficient of T in Q_n is $Q_{n-1}(0)^2 = 1$. Therefore Q_n is nonconstant and has a root in the algebraically closed field K . Such a root is nonzero because $Q_n(0) = 1$.

Let p be any prime. Choose a nonzero root c_p of Q_{p-1} . Then $P_p(c_p) = c_p Q_{p-1}(c_p) = 0$, so 0 is periodic under F_{c_p} with period dividing p . Its period is not 1, since $F_{c_p}(0) = c_p \neq 0$. Primality forces the minimal period to equal p . In particular F_{c_p} is PCF, and the parameters associated with different primes are distinct. The infinitude of primes gives infinitely many such parameters.

Finally, suppose F_c and F_d are affinely conjugate via $\phi(X) = aX + b$, where $a \neq 0$. The relation $\phi \circ F_c = F_d \circ \phi$ reads

$$a(X^2 + c) + b = (aX + b)^2 + d.$$

Comparing coefficients of X^2 yields $a = a^2$, hence $a = 1$. Comparing coefficients of X then yields $2ab = 0$, hence $b = 0$. The constant terms give $c = d$. Thus distinct parameters in the normalized family represent distinct affine-conjugacy classes. $\square$

Formalization and scope. The Lean file uses $K = \text{AlgebraicClosure}(\mathbb{Q})$ and proves the orbit-polynomial identity, existence of a nonzero root of every Q_n with $n \geq 1$, exact prime periods, infinitude of normalized PCF parameters, and affine-conjugacy rigidity. The decisive theorems are

- `TLMC7755.infinite_normalized_pcf_parameters`;
- `TLMC7755.normalized_affine_conjugate_iff`.

The axiom audit lists only `propext`, `Classical.choice`, and `Quot.sound`. We disprove the conjecture's first assertion; we make no claim about the separate height asymptotic.